ATHREYA KOTARAJARAMA

Senior Technical Specialist · SDK Quality Engineering · Automation Architecture · Platform Validation

krathreya@gmail.com | linkedin.com/in/athreya-k-r-75034167 | github.com/AthreyaAithal | Bengaluru, India

EXECUTIVE SUMMARY

Senior engineering leader with 11+ years building mission-critical **SDK quality engineering programs**, scalable **automation frameworks**, and **platform validation architectures** for large-scale consumer hardware ecosystems. Combines rare Clang-LLVM compiler depth with hands-on automation architecture experience and a leadership track record spanning technical direction, team guidance, and cross-regional engineering collaboration (India · Japan · USA). Proven ability to translate long-term quality requirements into maintainable, scalable engineering systems that raise validation maturity and accelerate delivery confidence.

11+	Years platform SDK QA	**100+**	QA cycles delivered on time	**300+**	Defects surfaced & escalated

LEADERSHIP & ARCHITECTURE HIGHLIGHTS

- ▸ Architected and led development of a large-scale internal Automation Test System (ATS) framework adopted across multiple SDK regression programs — improving execution reliability and cross-platform test coverage at scale.
- ▸ Designed a one-click MEAN stack DQA dashboard eliminating 18 hours of manual execution overhead per cycle — direct, measurable impact on team velocity and delivery confidence.
- ▸ Directed C++ standard library quality ownership (CPP20 additions, test refactoring, dinkum/plumhall/libcxx) — driving long-term maintainability and standards compliance across SDK library programs.
- ▸ Championed structure-aware fuzzing adoption in collaboration with US engineering teams, integrating libprotobuf-mutator into platform validation — extending defect detection beyond conventional API testing boundaries.
- ▸ Pioneered AI-assisted test generation proposals, positioning the quality engineering practice for next-generation automation and intelligent regression reduction.
- ▸ Led cross-regional engineering collaboration across India, Japan, and the United States — aligning quality strategy, technical direction, and execution planning with globally distributed stakeholder groups.

PROFESSIONAL EXPERIENCE

Senior Technical Specialist · Sony India Software Centre (SISC) · SARD *2023 – Present*

- Provide strategic technical direction for SDK quality programs; shape framework and automation architecture for multi-platform validation streams.
- Guide and mentor engineers on execution discipline, structured test strategy, and sound quality engineering practices.
- Drive engineering process improvements to increase validation maturity, execution reliability, and long-term maintainability of quality systems.

Technical Specialist · Sony India Software Centre (SISC) · SARD *2018 – 2023*

- Owned ATS framework architecture — full lifecycle from requirement analysis, design, and implementation through team adoption and iteration.
- Built internal productivity tools (test generator, spec finder, regression automation system, UI property fuzzer) that measurably improved team throughput.
- Owned C++ standard library test programs across dinkum, plumhall, and libcxx with CPP20 feature additions and systematic test refactoring.
- Implemented structure-aware fuzzing using libprotobuf-mutator in collaboration with US-based engineering teams.

Senior Software Engineer · Sony India Software Centre (SISC) · SARD *2015 – 2018*

- Designed and built a full-stack one-click automation dashboard (MEAN stack) eliminating 18 hours of manual DQA effort per cycle.

- Led SDK regression test execution and result analysis across two platform generations; developed Unicode and emoji test automation suites.
- Prepared and presented technical proposals for native-layer UI test approaches in collaboration with developers and backend engineering.

Software Engineer · Sony India Software Centre (SISC) · SARD *2013 – 2015*

- Contributed to compiler development using Clang-LLVM infrastructure — building low-level systems understanding that underpins all subsequent SDK and C/C++ quality work.
- Supported platform SDK regression test execution, quality analysis, and structured validation workflow development.

CORE EXPERTISE

Domains	SDK Quality Engineering · Automation Architecture · Platform Validation · Development QA · Software QA · Regression Engineering
Systems	Clang-LLVM Compiler Infrastructure · C++ Standard Library · C/C++ Ecosystem · Native Layer & API Testing · Structure-Aware Fuzzing
Tools	Python · JavaScript · C/C++ · MEAN Stack · Git · SVN · JIRA · Confluence · Shell Scripting
Leadership	Technical Direction · Team Mentoring · Cross-Regional Collaboration · Quality Strategy · Proposal & Feasibility Analysis · AI-Augmented Test Engineering

EDUCATION

MTech, Computer Science & Engineering (WILP) · BITS Pilani *2019*

B.E., Computer Science Engineering · NIE Mysore (Visvesvaraya Technological University) *2013*